JJ-B-00178c(DSA-DM)
29 September 1967
Interim Revision of
Fed. Spec. JJ-B-178b
19 April 1957

INTERIM FEDERAL SPECIFICATION

BELTS, ROUND, DENTAL ENGINE

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification JJ-B-178b.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers round, all-cord, dental engine belts suitable for power transferral from dental engines to dental handpieces.

1.2 Classification.

1.2.1 Sizes. Dental engine belts shall be furnished in the following sizes (lengths), as specified (see 6.1).

Size 1.	11 feet, 0 inch.
Size 2.	17 feet, 3 inches.
Size 3.	9 feet, 4 inches.
Size 4.	9 feet, 10 inches.
Size 5.	6 feet, 6 inches.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 3030

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SPECIFICATIONS

Federal

MMM-A-250	Adhesive, Water-Resistant, (For Sealing Fiberboard Boxes).
PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Setup.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing).

STANDARDS

Federal

FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications, required by activities outside the Federal Government for bidding purposes, are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D.C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

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STANDARDS

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Copies of Military standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The belts shall be fabricated from linen or semi-glazed cotton cord, composed of 3-ply or 4-ply number 16 thread, and suitable for the purpose intended.

3.2 Style and design. The belts shall be round, all-cord belts, suitable for transferring power from dental engines to dental handpieces. The cord shall be so braided and interwoven as to form a continuous unspliced round belt with a substantially uniform 0.1 inch diameter throughout its entire length. The design and fabrication shall be such as to minimize slipping of the belt on the pulleys. The belt shall also be free from any design imperfections which would create irregular operation of the activated handpiece.

3.3 Length. The belts shall be furnished in the lengths and tolerances shown in table I for the applicable size. Length shall be measured as specified in 4.4.1.

Table I. Length

Size	Length	Plus tolerance
1	11 feet, 0 inch	2 inches
2	17 feet, 3 inches	2 1/4 inches
3	9 feet, 4 inches	1 3/4 inches
4	9 feet, 10 inches	1 3/4 inches
5	6 feet, 6 inches	1 1/2 inches

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3.4 Breaking strength. The breaking strength shall be no less than 60 pounds, when tested as specified in 4.4.2.

3.5 Elongation. The elongation shall not exceed one percent of the original length of the belt, when tested as specified in 4.4.3.

3.6 Workmanship. Workmanship shall be first class throughout. The finished belts shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection level and acceptable quality levels (AQLs) shall be as shown in table II.

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Table II. Examination Criteria

Type defect	Inspection level	AQL (Percent defective)
Major	II	2.5
Minor	II	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. Test criteria shall be as indicated in table III. The unit of product shall be one complete belt for each test. The inspection level shall be S-2, and at least 5 units shall be tested regardless of lot size.

Table III. Test Criteria

Characteristic	Requirement	Test procedure	Requirement applicable to		Acceptance number
			Indiv. unit	Lot avg.	
Length	3.3	4.4.1	X		*
Breaking strength	3.4	4.4.2		X	Not appl.
Elongation	3.5	4.4.3	X		*

* The acceptance number shall be zero for all sample sizes.

4.3 Examination. The belts shall be examined to determine compliance with specification requirements. Examination shall be conducted in accordance with the classification of possible defects listed in table IV, but not restricted thereto.

Table IV. Classification of defects

Category	Defect
Major:	
101	Material of belt other than as specified.
102	Cord not composed of 3-ply or 4-ply number 16 thread.
103	Cord improperly or inadequately braided or interwoven.
104	Belt not round or not continuous.
105	Belt not free from the design imperfections specified.
Minor:	
201	Belts not free from defects which detract from their appearance.
202	Workmanship not first class throughout.
203	Diameter of belt not substantially uniform.
204	Packaging not as specified.
205	Packing not as specified.
206	Marking illegible, incorrect, or incomplete.

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4.4 Tests.

4.4.1 Length measurement. The length of the belt shall be measured under tension, one minute after application of a three-pound load. The belt may then be used for the elongation test specified in 4.4.3.

4.4.2 Breaking strength. The breaking strength of the belt shall be determined by using an inclination balance type machine with drum-type smooth grips no less than one inch in diameter. The pulling drum shall travel at a uniform rate of 12.0 ± 2.0 inches per minute under no load. The test shall be made at an ambient temperature of 70.0 ± 2.0 degrees Fahrenheit (° F.) and a relative humidity of 65.0 ± 2.0 percent. The distance between the centers of the drum shall be 10.0 ± 2.0 inches at the beginning of the test. At least 5 belts shall be selected for breaking strength, with the joint section centered between the grip. If the specimen breaks less than one-half inch from the point at which it touches the drum, or at any point touching the drum, the result shall be disregarded. After rupture of the specimen, the breaking force shall be read and the value recorded. The results for each lot or batch shall be averaged.

4.4.3 Elongation. The belt shall be measured in accordance with the requirements specified in 4.4.1., after which the entire uncut belt shall be subjected to a 25-pound tension for 24 hours. The tension shall then be released, and another measurement taken 24 hours later, in accordance with the requirements of 4.4.1. The difference between the second and first measurement, multiplied by 100 and divided by the first measurement, shall be the percentage of elongation.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. Each belt, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table V shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

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Table V. Packaging Quantities

Size	Unit package	Intermediate package	Exterior container
2	1 unit	12 units	288 units
1, 3, 4 and 5	1 unit	12 units	576 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors are acceptable. Closure of carton shall be adequate to prevent spilling of contents under normal handling.

5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2-inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type I, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant. Closure and strapping shall be in accordance with appendix of applicable box specification.

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5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic. Closure shall be in accordance with appendix of applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civil agencies. Shipments shall be marked in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Size required (see 1.2.1).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet Federal Government requirements.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: JJ-B-00178c(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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